

The diagram illustrates the relationship between three mathematical objects:

- A red horizontal rectangle representing a vector y . Above it is a horizontal arrow pointing right, labeled with the letter T .
- An equals sign $=$ in the center.
- A gray horizontal rectangle representing a vector $w^\top$. Above it is a horizontal arrow pointing right, labeled with the letter D .
- A blue square representing a matrix $X \in \mathbb{R}^{D \times T}$. Above it is a horizontal arrow pointing right, labeled with the letter T . To its left is a vertical arrow pointing down, labeled with the letter D .

The overall equation is:

$$y = w^\top X$$